UNIVERSITY NAME

Analysis IV — Final Exam

Year: 2023

Exercise 1:

Determine the nature (convergence/divergence) of the following numerical series:

$$1)\ u_n = \frac{n^n}{n!}$$
$$2)\ u_n = \log\left(1 + \frac{2}{n(n+1)}\right)$$
$$3)\ u_n = \sin\left(\frac{\pi}{2}\frac{an+1}{\sqrt{n^2+an+1}}\right)$$

Answer Area

1) Use Stirling's formula: $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$

   Then:

$$u_n = \frac{n^n}{n!} \sim \frac{n^n}{\sqrt{2\pi n}\left(\frac{n}{e}\right)^n} = \frac{e^n}{\sqrt{2\pi n}}$$

   Since $\lim_{n\to\infty} \frac{e^n}{\sqrt{2\pi n}} = +\infty$, the sequence diverges to $+\infty$.

2) First note that:

$$\frac{2}{n(n+1)} = 2\left(\frac{1}{n} - \frac{1}{n+1}\right)$$

   For the convergence of the series $\sum u_n$, use the asymptotic expansion:

$$\log(1+x) = x - \frac{x^2}{2} + o(x^2) \quad \text{as } x \to 0$$

   With $x = \frac{2}{n(n+1)} \sim \frac{2}{n^2}$:

$$u_n = \frac{2}{n(n+1)} - \frac{1}{2}\left(\frac{2}{n(n+1)}\right)^2 + o\left(\frac{1}{n^4}\right)$$

   The main term: $\frac{2}{n(n+1)} = 2\left(\frac{1}{n} - \frac{1}{n+1}\right)$

   So:

$$\sum_{n=1}^{N} \frac{2}{n(n+1)} = 2\sum_{n=1}^{N}\left(\frac{1}{n} - \frac{1}{n+1}\right) = 2\left(1 - \frac{1}{N+1}\right) \to 2$$

   The remaining terms form a convergent series since they are $O(1/n^4)$.

3) First analyze the argument:

$$\frac{an+1}{\sqrt{n^2+an+1}} = \frac{a+\frac{1}{n}}{\sqrt{1+\frac{a}{n}+\frac{1}{n^2}}}$$

As $n \to \infty$:

$$\frac{an+1}{\sqrt{n^2+an+1}} \to a$$

So:

$$\frac{\pi}{2}\frac{an+1}{\sqrt{n^2+an+1}} \to \frac{\pi a}{2}$$

And thus:

$$u_n \to \sin\left(\frac{\pi a}{2}\right)$$

Exercise 2:

For all $n \in \mathbb{N}^*$, define the function:

$$f_n : \mathbb{R} \longrightarrow \mathbb{R}, \quad x \mapsto \frac{n^\alpha x}{1+n^2x^2}$$

When the series $\sum f_n(x)$ converges, denote $f_\alpha(x) = \sum_{n=1}^{\infty} f_n(x)$.

1. Determine, according to the values of α, the domain D_α of definition of f_α.

   From now on, assume $\alpha < 1$.

2. (a) Show that for all $x \in \mathbb{R}_+^*$,

   $$\frac{x}{1+x^2}n^{\alpha-2} \le f_n(x) \le \frac{1}{x}n^{\alpha-2}.$$

   (b) Show that there exists $A > 0$ (depending on α) such that

   $$\forall x \in \mathbb{R}_+^*, \quad \frac{Ax}{1+x^2} \le f_\alpha(x) \le \frac{A}{x}.$$

   Deduce an equivalent of f_α at ∞.

3. If $\alpha < 0$, show that

   $$\int_0^1 f_\alpha(t)\,dt = \sum_{n=1}^{\infty} \frac{1}{2}n^{\alpha-2}\log(1+n^2).$$

4. Assume $0 \le \alpha < 1$.

   (a) Show that f_α is continuous on $\mathbb{R}^+$.

(b) Show that for all $N \in \mathbb{N}^*$

$$f_\alpha\left(\frac{1}{N}\right) \geq N \sum_{n=1}^{N} \frac{n^\alpha}{N^2 + n^2}$$

Deduce the discontinuity of f_α at 0.

(c) Does the series $\sum f_n$ converge uniformly on $\mathbb{R}_+$?

5. (Optional) Show that f_α is differentiable on $(0, +\infty)$ and decreasing on $[1, +\infty)$.

Answer Area

1. Domain D_α:

For $x = 0$: $f_n(0) = 0$ for all n, so the series converges to 0.

For $x \neq 0$: As $n \to \infty$,

$$|f_n(x)| \sim \frac{n^\alpha |x|}{n^2 x^2} = \frac{1}{|x|} n^{\alpha - 2}$$

By the p-series test, $\sum n^{\alpha-2}$ converges if and only if $\alpha - 2 < -1$, i.e., $\alpha < 1$.

Also, for fixed $x \neq 0$, $f_n(x) = O(n^{\alpha-2})$, so by comparison:

- If $\alpha < 1$: $\sum |f_n(x)|$ converges for all $x \in \mathbb{R}$.
- If $\alpha = 1$: $f_n(x) \sim \frac{1}{n|x|}$, and $\sum \frac{1}{n}$ diverges.
- If $\alpha > 1$: Diverges even faster.

2. Assuming $\alpha < 1$:

(a) For $x > 0$:

Lower bound: Since $1 + n^2 x^2 \geq 1$,

$$f_n(x) = \frac{n^\alpha x}{1 + n^2 x^2} \leq n^\alpha x$$

Actually, we need $n^{\alpha-2}$ bound. Better approach:

$$f_n(x) = \frac{x}{1 + x^2} \cdot \frac{n^\alpha (1 + x^2)}{1 + n^2 x^2}$$

Consider $g(t) = \frac{1+t^2}{1+n^2 t^2}$ for fixed n and $t = x > 0$... Wait, the given inequality is:

$$\frac{x}{1 + x^2} n^{\alpha-2} \leq f_n(x) \leq \frac{1}{x} n^{\alpha-2}$$

Check: $f_n(x) = \frac{n^\alpha x}{1+n^2x^2} = \frac{1}{x} \cdot \frac{n^\alpha x^2}{1+n^2x^2} \leq \frac{1}{x} \cdot \frac{n^\alpha}{n^2} = \frac{1}{x} n^{\alpha-2}$

For the lower bound: $f_n(x) \geq \frac{n^\alpha x}{1+x^2} \cdot \frac{1}{n^2}$ when $n^2x^2 \leq 1$? Actually:

$f_n(x) = \frac{x}{1+x^2} \cdot n^{\alpha-2} \cdot \frac{n^2(1+x^2)}{1+n^2x^2} \geq \frac{x}{1+x^2} n^{\alpha-2}$ if $\frac{n^2(1+x^2)}{1+n^2x^2} \geq 1$

But $\frac{n^2(1+x^2)}{1+n^2x^2} = \frac{n^2+n^2x^2}{1+n^2x^2} \geq 1$ since $n^2 \geq 1$.

(b) Sum the inequalities:
Let $A = \sum_{n=1}^{\infty} n^{\alpha-2}$ (converges since $\alpha - 2 < -1$ when $\alpha < 1$).
Then:

$$\frac{x}{1+x^2} A \leq f_\alpha(x) \leq \frac{1}{x} A$$

As $x \to \infty$: $\frac{x}{1+x^2} \sim \frac{1}{x}$, so $f_\alpha(x) \sim \frac{A}{x}$.

3. For $\alpha < 0$:
Want to show: $\int_0^1 f_\alpha(t)dt = \sum_{n=1}^{\infty} \frac{1}{2} n^{\alpha-2} \log(1+n^2)$
Compute the integral:

$$\int_0^1 f_n(t)dt = \int_0^1 \frac{n^\alpha t}{1+n^2t^2} dt$$

Let $u = 1 + n^2t^2$, then $du = 2n^2 t dt$, so $tdt = \frac{du}{2n^2}$.
When $t = 0$: $u = 1$; when $t = 1$: $u = 1 + n^2$.
So:

$$\int_0^1 f_n(t)dt = n^\alpha \int_1^{1+n^2} \frac{1}{u} \cdot \frac{du}{2n^2} = \frac{n^{\alpha-2}}{2} \int_1^{1+n^2} \frac{du}{u} = \frac{1}{2} n^{\alpha-2} \log(1+n^2)$$

Now interchange sum and integral (justified by uniform convergence or monotone convergence since all terms are positive):

$$\int_0^1 f_\alpha(t)dt = \int_0^1 \sum_{n=1}^{\infty} f_n(t)dt = \sum_{n=1}^{\infty} \int_0^1 f_n(t)dt = \sum_{n=1}^{\infty} \frac{1}{2} n^{\alpha-2} \log(1+n^2)$$

4. For $0 \leq \alpha < 1$:

(a) Continuity on $\mathbb{R}^+$:
For $x > 0$, each f_n is continuous and we have uniform convergence on $[x_0, \infty)$ for any $x_0 > 0$ since:

$$|f_n(x)| \leq \frac{1}{x_0} n^{\alpha-2} \quad \text{for } x \geq x_0$$

and $\sum n^{\alpha-2}$ converges.
At $x = 0$: $f_n(0) = 0$ and $f_\alpha(0) = 0$. Need to check continuity at 0.

(b) For $x = \frac{1}{N}$:

$$f_\alpha\left(\frac{1}{N}\right) = \sum_{n=1}^{\infty} \frac{n^\alpha \cdot \frac{1}{N}}{1 + n^2 \cdot \frac{1}{N^2}} = N \sum_{n=1}^{\infty} \frac{n^\alpha}{N^2 + n^2}$$

So:

$$f_\alpha\left(\frac{1}{N}\right) \geq N \sum_{n=1}^{N} \frac{n^\alpha}{N^2 + n^2} \geq N \sum_{n=1}^{N} \frac{n^\alpha}{N^2 + N^2} = \frac{1}{2N} \sum_{n=1}^{N} n^\alpha$$

For $0 \leq \alpha < 1$, $\sum_{n=1}^{N} n^\alpha \sim \frac{N^{\alpha+1}}{\alpha+1}$.
So:

$$f_\alpha\left(\frac{1}{N}\right) \gtrsim \frac{1}{2N} \cdot \frac{N^{\alpha+1}}{\alpha+1} = \frac{1}{2(\alpha+1)} N^\alpha$$

Since $\alpha \geq 0$, $N^\alpha \geq 1$, so $f_\alpha\left(\frac{1}{N}\right)$ doesn't tend to 0 as $N \to \infty$.
But $f_\alpha(0) = 0$, so f_α is discontinuous at 0.

(c) Uniform convergence on $\mathbb{R}_+$?
No, because if the series converged uniformly on $\mathbb{R}_+$, the sum would be continuous at 0 (as each f_n is continuous), but we just showed discontinuity at 0.

5. Differentiability and monotonicity:
For $x > 0$:

$$f_n'(x) = \frac{n^\alpha(1+n^2x^2) - n^\alpha x \cdot 2n^2 x}{(1+n^2x^2)^2} = \frac{n^\alpha(1-n^2x^2)}{(1+n^2x^2)^2}$$

For $x \geq 1$, $f_n'(x) \leq 0$ since $1 - n^2x^2 \leq 0$.
Check uniform convergence of $\sum f_n'(x)$ on $[1, \infty)$:

$$|f_n'(x)| = \frac{n^\alpha|1-n^2x^2|}{(1+n^2x^2)^2} \leq \frac{n^\alpha \cdot n^2x^2}{(n^2x^2)^2} = \frac{n^{\alpha+2}x^2}{n^4x^4} = \frac{n^{\alpha-2}}{x^2} \leq n^{\alpha-2}$$

Since $\sum n^{\alpha-2}$ converges, $\sum f_n'$ converges uniformly on $[1, \infty)$, so f_α is differentiable on $(0, \infty)$ and decreasing on $[1, \infty)$.

Exercise 3:

Consider the differential equation:

$$4xy''(x) + 2y'(x) - y(x) = 0 \quad (E)$$

Let $\sum a_n x^n$ be a power series with radius of convergence R and sum f.

1. Justify that f is of class C^2 on $(-R, R)$. Deduce, for $x \in (-R, R)$, an expression of $f'(x)$ and $f''(x)$.

2. Show that f is a solution of (E) if and only if

$$a_{n+1} = \frac{a_n}{(2n+1)(2n+2)}, \quad \forall n \in \mathbb{N}.$$

3. Express a_n in terms of n and a_0. Determine the radius of convergence of the power series $\sum a_n x^n$.

4. Expand into a power series the function defined by:

$$\varphi(x) = \begin{cases} \cosh(\sqrt{x}) & \text{if } x > 0 \\ 1 & \text{if } x = 0 \\ \cos(\sqrt{-x}) & \text{if } x < 0 \end{cases}$$

Answer Area

1. If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ with radius R, then:
   - f is C^∞ on $(-R, R)$
   - $f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$
   - $f''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$
2. Substitute into (E):

$$4xf''(x) + 2f'(x) - f(x) = \sum_{n=0}^{\infty} [4(n+2)(n+1)a_{n+2}x^{n+1} + 2(n+1)a_{n+1}x^n - a_n x^n]$$

Shift index in first sum: let $m = n+1$, then $n = m-1$:

$$= \sum_{m=1}^{\infty} 4(m+1)m a_{m+1} x^m + \sum_{n=0}^{\infty} [2(n+1)a_{n+1} - a_n] x^n$$

Combine terms for $n \geq 1$:

$$= (2a_1 - a_0) + \sum_{n=1}^{\infty} [4(n+1)n a_{n+1} + 2(n+1)a_{n+1} - a_n] x^n$$

For this to be 0 for all x in $(-R, R)$, all coefficients must be 0:

$$2a_1 - a_0 = 0 \quad \Rightarrow \quad a_1 = \frac{a_0}{2}$$
$$4(n+1)n a_{n+1} + 2(n+1)a_{n+1} - a_n = 0 \quad \text{for } n \geq 1$$

Simplify recurrence for $n \geq 1$:

$$(n+1)[4n+2]a_{n+1} = a_n \quad \Rightarrow \quad (n+1)(4n+2)a_{n+1} = a_n$$

But wait, the problem states: $a_{n+1} = \frac{a_n}{(2n+1)(2n+2)}$

Check: $(n+1)(4n+2) = 2(n+1)(2n+1) = 2(2n+1)(n+1)$

So: $a_{n+1} = \frac{a_n}{2(2n+1)(n+1)} = \frac{a_n}{(2n+1)(2n+2)}$

Also check for $n = 0$: $a_1 = \frac{a_0}{(2\cdot 0+1)(2\cdot 0+2)} = \frac{a_0}{1\cdot 2} = \frac{a_0}{2}$

So the recurrence holds for all $n \geq 0$.

3. Express a_n in terms of a_0:

   For $n = 0$: $a_0 = a_0$

   For $n = 1$: $a_1 = \frac{a_0}{1 \cdot 2} = \frac{a_0}{2!}$

   For $n = 2$: $a_2 = \frac{a_1}{3 \cdot 4} = \frac{a_0}{2! \cdot 3 \cdot 4} = \frac{a_0}{4!}$

   For $n = 3$: $a_3 = \frac{a_2}{5 \cdot 6} = \frac{a_0}{4! \cdot 5 \cdot 6} = \frac{a_0}{6!}$

   Pattern: $a_n = \frac{a_0}{(2n)!}$

   Check by induction: Assume $a_n = \frac{a_0}{(2n)!}$, then:

   $$a_{n+1} = \frac{a_n}{(2n+1)(2n+2)} = \frac{a_0}{(2n)!} \cdot \frac{1}{(2n+1)(2n+2)} = \frac{a_0}{(2n+2)!}$$

   So $f(x) = a_0 \sum_{n=0}^{\infty} \frac{x^n}{(2n)!}$

   Radius of convergence: Use ratio test:

   $$\left| \frac{a_{n+1} x^{n+1}}{a_n x^n} \right| = \left| \frac{x}{(2n+1)(2n+2)} \right| \to 0 \quad \text{as } n \to \infty$$

   So $R = \infty$.

4. The function $\varphi(x)$:

   For $x > 0$: $\cosh(\sqrt{x}) = \sum_{n=0}^{\infty} \frac{(\sqrt{x})^{2n}}{(2n)!} = \sum_{n=0}^{\infty} \frac{x^n}{(2n)!}$

   For $x < 0$: $\cos(\sqrt{-x}) = \sum_{n=0}^{\infty} (-1)^n \frac{(\sqrt{-x})^{2n}}{(2n)!} = \sum_{n=0}^{\infty} (-1)^n \frac{(-x)^n}{(2n)!} = \sum_{n=0}^{\infty} \frac{x^n}{(2n)!}$

   For $x = 0$: $\varphi(0) = 1 = \sum_{n=0}^{\infty} \frac{0^n}{(2n)!} = 1$

   So in all cases: $\varphi(x) = \sum_{n=0}^{\infty} \frac{x^n}{(2n)!}$

   This is exactly the solution of (E) with $a_0 = 1$.